# FOUNDATIONS OF GENERATIVE AI (ICT4403)

*Assignment 2: Autoregressive Text Generation System — Comparative Study*

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| **Assignment Title** | Assignment 2 - Autoregressive Text Generation System |
| **System Title** | Comparative Word-Level Autoregressive Text Generation on Lewis Carroll's Alice in Wonderland using Simple RNN, BiRNN, LSTM, GRU |
| **Dataset & Source** | Alice's Adventures in Wonderland (Lewis Carroll) \| https://www.gutenberg.org/files/11/11-0.txt |
| **GitHub Repository** | https://github.com/Saksham-Gupta-GH/fgai-autoregressive-text-generation-alice |

## 1. System Overview & Problem Statement

Four recurrent neural language models (Simple RNN, BiRNN with forward-backward final state concatenation [h_n[-2], h_n[-1]], LSTM, and GRU) were trained under strictly controlled conditions on Alice's Adventures in Wonderland. To eliminate data leakage, the corpus is partitioned chronologically (80% Train, 10% Val, 10% Test) and the vocabulary (1,300 tokens, min_freq=2) is fitted exclusively on training tokens. Context sequence length is 6 tokens. Held-out test set has an 11.17% <unk> token rate.

## 2. Quantitative Evaluation Summary (Best Epoch & Test Set)

| Architecture | Parameters | Best Ep | Val Loss | Val PPL | Val Top-1 | Test Loss | Test Top-1 | Test Top-5 |
|---|---|---|---|---|---|---|---|---|
| **Simple RNN** | 798,484 | 4 | 4.0741 | 58.80 | 19.73% | 4.1584 | 20.80% | 45.28% |
| **BiRNN** | 1,262,868 | 2 | 4.2804 | 72.27 | 18.66% | 4.3192 | 18.47% | 42.26% |
| **LSTM** | 1,193,236 | 5 | **3.9903** | **54.07** | **21.87%** | **4.0576** | **23.29%** | **47.23%** |
| **GRU** | 1,061,652 | 3 | 4.0353 | 56.56 | 20.55% | 4.0900 | 22.53% | 46.51% |

## 3. Cross-Model Temperature Analysis (Averaged Over Models and Seed Prompts, n = 3 per cell)

| T | Distinct-1 | Distinct-2 | Trigram Repetition | Longest Loop | Empirical Quality Observation |
|---|---|---|---|---|---|
| **0.2** | 0.517 | 0.674 | 23.7% | 0–6 words | Strict syntax; repetitive looping traps across models (GRU 52.5%) |
| **0.5** | 0.678 | 0.900 | 3.0% | 0 words | Optimal empirical trade-off: high diversity with fluent story dialogue; LSTM and GRU comparable |
| **1.0** | 0.855 | 0.992 | 0.0% | 0 words | High lexical richness; slight syntactic drift across longer clauses |
| **1.5** | 0.916 | 1.000 | 0.0% | 0 words | High entropy; syntactic collapse into ungrammatical token sequences |

## 4. Conclusion & Model Comparison

• **LSTM vs. GRU & UNK Nuance:** LSTM leads across next-word prediction metrics (Val Loss 3.9903, PPL 54.07, Test Top-1 23.29%), with GRU performing closely behind (Val Loss 4.0353, Test Top-1 22.53%) with 11% fewer parameters. Test targets include 11.17% <unk> tokens due to train-only vocabulary fitting.
• **Qualitative Consistency & Seed Origin:** Seed 1 comes from the training text, Seed 2 from held-out test text, and Seed 3 from validation text. Consistency was assessed qualitatively: at T=0.5, LSTM and GRU more frequently generate Carroll-style dialogue tags ('the march hare said'), though all models drift across multi-clause sequences without sustaining a persistent scene.
• **Single-Seed Limitation & BiRNN:** SEED=42 ensures strict comparability; the 0.045 loss gap between LSTM and GRU falls within single-run variance margins. BiRNN's forward-backward state concatenation feeds complete context, but its 1,262,868 parameters overfit by epoch 2 (val loss 4.2804).
• **Temperature Sampling:** T = 0.5 provides the optimal trade-off between lexical diversity (Distinct-1: 0.60–0.73) and phrase stability, achieving minimal trigram repetition (LSTM 5.0%, GRU 0.0%, BiRNN 1.0%, Simple RNN 6.1%).